

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- Client - DriverPass
- Purpose - Deliver a business management system
 - Deliver educational material to users
 - Allow users to book on-the-road training appointments
 - Let DriverPass employees manage user accounts and appointments
 - Allow DrivePass boss to access data from anywhere

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- Assist customers in passing their driving test
- Provide practice tests
- Allow for easy access to DriverPass on-the-road training

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- DriverPass Customers
 - book a reservation for driving lessons
 - day and time
 - Pickup and drop off location
 - purchase a package

- x hours of on-the-road training with a driver
 - possible in-person lesson
 - possible access to online classes including materials, tests, and practice tests
- “forgot password” reset system
- DriverPass
 - access and download data from PC or mobile
 - employee account rights and roles management
 - Boss - online data access and employee account management including password resets and rights editing
 - IT Officer - system maintenance and modification
 - User Manager - create/modify user accounts, create/modify appointments for users
 - Driver - view appointment details/ schedule
 - reservation activity record
 - track changes, cancellations, and who modified it
 - print activity report
 - book a reservation for a customer/modify appointments
 - booked reservations match a customer with a driver and a car
 - ability to disable/enable packages available to customers
 - user account management/creation
 - user account details
 - first/last name, address, phone number, state, credit card (number, exp. date, security code), drop-off/pickup location
 - DMV update notifications
 - alert with new rules, policies, or sample questions
 - cloud based web site
 - third party security and backups
- Primary User Interface
 - online test progress of customer
 - display tests in progress and completed
 - name, time taken, score, status(taken, in progress, failed, passed)
 - driver notes
 - display comments from drivers in a table format
 - lesson time, start hour, end hour, driver comments
 - Information
 - first/last name, address, city, state, zip, phone, email...
 - special needs
 - driver photo

- student photo

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- Environments: web and mobile with offline support
- Update when users submit changes (reservation modifications, profile changes, payments made)
- No speed specified. (Average website response time: 100-200ms)

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- Platform independent (web-based)
- Fully cloud based backend
- Record of changes (who made what changes and when)
- Viewable/printable activity report
- Data reports that can be edited/downloaded/uploaded (to make changes)

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Users
 - Roles separate system rights and level of access
 - Distinguishing ID number with first and last names as secondary identification
 - Login with case sensitive password and user name
- Automated Password reset
- Admin alert when DMV updates rules, policies, or sample questions

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

Non-code changes:

- Admin

- Package disabling and enabling
 - Full control to modify employee accounts such as password reset or restrict access
- Employee
 - Add/remove/modify customer accounts
 - Fully edit reservations
- Customers
 - Submit/modify reservations

Code Changes:

- Developer
 - Add/remove packages available to customers

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a “brute force” hacking attempt? What happens if the user forgets their password?

- User login: case sensitive name and password
 - automated customer password reset if they forgot it
- Security will be handled by a third party over cloud services

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with “The system shall . . .” For example, one functional requirement might be, “The system shall validate user credentials when logging in.”

- The system shall provide data to the boss to download and edit elsewhere
- The system shall allow employees to add and edit customer accounts
- The system shall allow employees to change appointment details
- The system shall allow customers to purchase packages with their recorded payment information
- The system shall allow customers to submit and change appointments
- The system shall match an available driver to a car and match them to a customer when an appointment is made
- The system shall provide customers with access to test materials and access to the online classes if the package they purchased includes it
- The system shall validate user credentials when logging in
- The system shall record when reservations are made or modified and who modified it last
- The system shall allow admins to change and block employee accounts
- The system shall defer security and payment processing to third parties for validation
- The system shall alert admins when the DMV changes any rules, policies, or test practice questions

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- Mobile/browser support
- Customer and employee access through the log in portal
- Customers see their account info (name, full address, phone, email) , online test progress, driver notes, special needs, driver photo, and student photo
 - Test progress displays tests taken by customer, the test's name, score, time taken, and progress in the form of words (in progress, failed, passed, not taken)
 - Driver notes are in the form of a table and show lesson time, start/end hour, and driver comments
- Employees can view and change customer account info after an account lookup
- Customers or employees will use this interface to input customer information
- Customers will also see a contact page to contact the company
- Employees will see a way to contact the customer

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- More interfaces will be designed later to handle events such as online tests
- Third party services will be found and used to handle such things as back end security since the owner wants to focus on business with minimal technical overhead
- Drivers are managed outside of the system or simply emailed driving lesson details as little was specified about them in regards to this system
- The boss will have the same rights as his IT officer/admin
- All users have a web browsing capable device (PC/smart phone)
- Customers must first sign up over the phone before gaining access to the website

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Budget. Relying on third parties for services incurs costs that may be a lot depending on scale.
- Third party services will limit the capabilities of the system such as metering traffic or data limits.
- Being entirely cloud based also limits a degree of control as servers and databases can only be controlled remotely and through APIs.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

Driver Pass

